

Jucheng Shen

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EDUCATION

Rice University

Expected May 2027

B.A. in Computer Science, Math, Econ, GPA – 3.7/4.0

- Relevant Coursework: Algorithms and Data Structures, Discrete Mathematics, Computer Systems, Honors Linear Algebra, Real Analysis, Probability and Statistics, Honors ODE, Differential Geometry, Computer Vision

EXPERIENCE

Research Intern | California Institute of Technology, advised by Prof. Anima Anandkumar

Feb 2026 – Present

- Investigating neural operator learning on continuous language representations for text generation.

Research Intern | Rice University, advised by Prof. Anastasios Kyrillidis

Jan 2026 – May 2026

- Proposed *Asymmetric Input Recurrence (AIR)*, a shared-weight recurrent Transformer that induces proposal-like z_H and scratchpad-like z_L roles through asymmetric input injection.
- Designed decoded-state, input-ablation, freeze, and attention analyses showing local z_L updates, global z_H updates, and AIR variants that match or exceed HRM with half the parameters.

Research Intern | Princeton University, advised by Prof. Zhuang Liu

Nov 2025 – Feb 2026

- Investigating data filtering and dataset merging strategies for large-scale text-to-image models, with a focus on improving generalization and training efficiency.
- Studying the role of image-text alignment by analyzing how VAE feature-caption consistency influences representation learning in open text-to-image foundation models.

Research Intern | UT Austin & Intel Labs, advised by Prof. Atlas Wang & Dr. Souvik Kundu

May 2025 – Sep 2025

- Proposed *One-Shot Dynamic Thresholding (OSDT)*, a training-free adaptive decoding method for diffusion LLMs achieving 24–50% higher throughput without accuracy loss.
- Led a collaboration with Intel Labs, proposing *CadLLM*, a unified adaptive decoding system that jointly tunes step size, block size, vocabulary, and thresholds, achieving up to $2.3\times$ speedup on diffusion LLMs.

Research Intern | Rice University, advised by Prof. Yuke Wang

Apr 2025 – Aug 2025

- Implemented *SuperGen*, a novel framework enabling generation of ultra-high resolution (2–4K) videos with limited GPU resources, on two baseline model architectures.
- Conducted end-to-end and ablation experiments, assisted with figure production and paper writing for publication.

PUBLICATIONS

One Model, Two Roles: Emergent Specialization in a Shared Recurrent Transformer. arXiv preprint 2026. **Shen, J.**, Su, W., Kyrillidis, A. [arXiv:2605.17811](https://arxiv.org/abs/2605.17811).

Improving the Throughput of Diffusion-based Large Language Models via a Training-Free Confidence-Aware Calibration. Findings of ACL 2026. **Shen, J.**, Sarkar, G., Ro, Y., Nittur Sridhar, S., Wang, Z., Akella, A., Kundu, S. [arXiv:2512.07173](https://arxiv.org/abs/2512.07173).

Beyond Static Cutoffs: One-Shot Dynamic Thresholding for Diffusion Language Models. NeurIPS 2025 Efficient Reasoning Workshop. **Shen, J.**, Ro, Y. [arXiv:2511.02077](https://arxiv.org/abs/2511.02077).

SuperGen: An Efficient Ultra-High-Resolution Video Generation System with Sketching and Tiling. Preprint 2025. Ye, F., Zhao, Z., Mu, Y., **Shen, J.**, Li, R., Wang, K., Sun, D., Agarwal, S., Lee, M., Cao, T., Akella, A., Krishnamurthy, A., Ng, T. S. E., Tu, Z., Wang, Y. [arXiv:2508.17756](https://arxiv.org/abs/2508.17756).

PROJECTS

Board Member (Tracks and Workshops) | HackRice

Apr 2025 – Oct 2025

- Hosted 2 technical workshops for 100+ attendees; ran Q&A and live coding demos.
- Co-designed and shipped the official starter code used by 500+ hackers ([repo](#)).
- Designed competition tracks and judging criteria for HackRice 15.

HONORS & AWARDS

Frank Liu Jr. Prize for Creative Innovations in Music, Fashion, and the Arts in Napier Rice Launch Challenge ([link](#)).

TECHNICAL SKILLS

Tools & Frameworks: Python, Numpy, PyTorch, JAX, Bash/Shell scripting, HPC/Slurm, C, Java

Skills: LLM evaluation & benchmarking, large-scale experiment management, academic writing